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| CONTACT INFORMATION     | Email: <a href="mailto:zhang.boyuan@hotmail.com">zhang.boyuan@hotmail.com</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Personal webpage: <a href="https://boyuan-zhang.github.io">boyuan-zhang.github.io</a> |
| RESEARCH FIELDS         | Econometrics, Time Series, Machine Learning, Applied Macroeconomics, Climate Econometrics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                       |
| CURRENT POSITION        | Amazon.com, Inc.<br>Economist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Bellevue, WA<br>2024 - Present                                                        |
| PREVIOUS POSITION       | Amazon.com, Inc.<br>Postdoctoral Scientist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Bellevue, WA<br>2023 - 2024                                                           |
| EDUCATION               | University of Pennsylvania<br>Ph.D. in Economics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Philadelphia, PA<br>2017 - 2023                                                       |
|                         | University of Illinois at Urbana-Champaign<br>M.S. in Statistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Champaign, IL<br>2014 - 2016                                                          |
|                         | Renmin University of China<br>B.S. in Physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Beijing, China<br>2010 - 2014                                                         |
| PUBLICATIONS            | <ol style="list-style-type: none"> <li>1. "On the Aggregation of Probability Assessments: Regularized Mixtures of Predictive Densities for Eurozone Inflation and Real Interest Rates," <i>Journal of Econometrics</i>, 237, 105321, 2023. With F. X. Diebold and M. Shin.</li> <li>2. "When Will Arctic Sea Ice Disappear? Projections of Area, Extent, Thickness, and Volume," <i>Journal of Econometrics</i>, 236, 105479, 2023. With F. X. Diebold, P. Goulet Coulombe, M. Göbel, and G. Rudebusch.</li> <li>3. "Optimal Combination of Arctic Sea Ice Extent Measures: A Dynamic Factor Modeling Approach," <i>International Journal of Forecasting</i>, 37, 1509-1519, 2021. With F. X. Diebold, P. Goulet Coulombe, M. Göbel, and G. Rudebusch.</li> <li>4. "Measuring International Uncertainty: the Case of Korea," <i>Economics Letters</i>, 162, 22-26, 2018. With M. Shin, M. Zhong, and D. Lee.</li> </ol> |                                                                                       |
| WORKING PAPERS          | <ol style="list-style-type: none"> <li>5. "Incorporating Prior Knowledge of Latent Group Structure in Panel Data Models." Working paper at <a href="https://arxiv.org/abs/2211.16714">arXiv:2211.16714</a>.</li> <li>6. "Bayesian Estimation of Sparsely Heterogeneous Panel Models," with H. R. Moon and F. Schorfheide. Working paper at <a href="https://arxiv.org/abs/2310.13785">arXiv:2310.13785</a>. R&amp;R, <i>Journal of Applied Econometrics</i>.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                       |
| WORK IN PROGRESS        | <ol style="list-style-type: none"> <li>7. "Disentangling Local and Aggregate Income Effects on Consumption: Evidence from U.S. States," with D. Giannone, M. Lenze, S. Li.</li> <li>8. "Estimate Nonlinear VARs using Neural Network Models," with P. Goulet Coulombe.</li> <li>9. "Early Warning Indicator of Banking Crises via Factor-Augmented VAR Models."</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                       |
| PROFESSIONAL ACTIVITIES | Referee: <i>Journal of Econometrics</i> , <i>Journal of Applied Econometrics</i> , <i>International Journal of Forecasting</i> , <i>Econometric Reviews</i> , <i>Journal of the Royal Statistical Society: Series C</i> , <i>Economics and Business Letters</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |

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| CONFERENCE &<br>SEMINAR<br>PRESENTATIONS | 2025: Conference on Real-Time Data Analysis, Methods, and Applications (Czech National Bank), 3rd Vienna Workshop on Economic Forecasting (ISF)<br>2024: International Symposium of Forecasting<br>2023: Rutgers University–New Brunswick, SEA Annual Meeting<br>2022: Young Scholars Conference on Machine Learning in Economics and Finance (Philadelphia Fed), NBER-NSF SBIES Conference (WUSTL), ES North American Summer Meeting, ES Asian Meeting, ES Australasia Meeting, IAAE Annual Conference, International Symposium of Forecasting, ASSA Annual Meeting (IAAE Invited Session)<br>2021: Annual SoFiE Conference, IAAE Annual Conference, CEA Conference<br>2019: 25th International Conference on Computing in Economics and Finance |                                                                                                                                                                                                                                                          |
| TEACHING<br>EXPERIENCE                   | University of Pennsylvania<br>Recitation Instructor<br>Econ 002: Introductory Macroeconomics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fall 2018, Spring 2019                                                                                                                                                                                                                                   |
| FELLOWSHIPS,<br>AWARDS, AND<br>GRANDS    | University Fellowship, University of Pennsylvania<br>Outstanding Graduate, Renmin University of China<br>China National Scholarship, Renmin University of China                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2017 - 2018<br>2014<br>2013                                                                                                                                                                                                                              |
| PROFESSIONAL<br>EXPERIENCE               | Economist Intern<br>Amazon.com, Inc., Seattle, WA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | May 2021 - Aug 2021                                                                                                                                                                                                                                      |
|                                          | Data Analyst Intern<br>Anheuser-Busch InBev, Inc., Champaign, IL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | May 2015 - Feb 2016                                                                                                                                                                                                                                      |
| COMPUTING<br>SKILLS                      | Proficiency in Python, R, Matlab, and Julia; knowledge in STATA and SAS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                          |
| PERSONAL<br>INFORMATION                  | Date of Birth: April 15th, 1992<br>Citizenship: China (F-1 Visa)<br>Languages: Chinese (native), English (fluent).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                          |
| REFERENCE                                | Francis X. Diebold (Co-advisor)<br>Department of Economics<br>University of Pennsylvania<br>133 South 36th Street, Office 607<br>Philadelphia, PA, 19104<br>Phone: 215-898-1507<br>E-mail: <a href="mailto:fdiebold@econ.upenn.edu">fdiebold@econ.upenn.edu</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Frank Schorfheide (Co-advisor)<br>Department of Economics<br>University of Pennsylvania<br>133 South 36th Street, Office 621<br>Philadelphia, PA, 19104<br>Phone: 215-898-8486<br>E-mail: <a href="mailto:schorf@ssc.upenn.edu">schorf@ssc.upenn.edu</a> |
|                                          | Xu Cheng<br>Department of Economics<br>University of Pennsylvania<br>133 South 36th Street, Office 620<br>Philadelphia, PA, 19104<br>Phone: 215-898-6775<br>E-mail: <a href="mailto:xucheng@econ.upenn.edu">xucheng@econ.upenn.edu</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Minchul Shin<br>Federal Reserve Bank of Philadelphia<br>Ten Independence Mall,<br>Philadelphia, PA, 19106<br>Phone: 215-574-6079<br>E-mail: <a href="mailto:minchul.shin@phil.frb.org">minchul.shin@phil.frb.org</a>                                     |